IDR as Insurance: Employment Effects of Income-Driven Repayment Expansion

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Abstract

Moral hazard arguments suggest that Income-driven repayment (IDR) plans for student loans weaken work incentives by lowering the return to earnings. I test this in the United States (US) context using two cohorts of federal student loan borrowers and find the opposite—the 2009 IBR expansion increased employment probability by 2.1 percentage points, with effects concentrated among lower-achieving, default-risk borrowers consistent with IDR operating as insurance against non-payment. Contrary to moral hazard predictions, hours worked and wages did not decline on average, a result I attribute to the absence of salient income thresholds in US IDR design. Black borrowers experienced 5–6 percent wage gains following both the 2009 and 2015 expansions.

Keywords: Student loans, Income-driven repayment, Labor supply, Higher education
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1. Introduction

A central concern of income-driven repayment (IDR) plans for student loans is that they can reduce work incentives. By tying monthly payments to earnings, IDR effectively lowers the net return to an extra hour of work for borrowers above the income allowance, like an implicit marginal tax that could discourage labor supply. Recent evidence from Australia finds that borrowers bunch just below salient repayment thresholds to reduce their income-contingent payments, a clear behavioral response to the program's design (de Silva, 2025). If the same moral hazard exists in the United States, then the rapid expansion of IDR over the past several years from 5 percent of borrowers in 2012 to over 33 percent by 2020 may have quietly eroded the work incentives for thousands of college-educated borrowers.

The United States presents a sharp institutional contrast. Unlike the Australian system, US IDR plans do not feature salient income thresholds that create notches in the budget set. Payments are designed to taper smoothly with earnings, enrollment requires active opt-in, and borrowers near the poverty allowance have typically owed nothing. For the large share of borrowers at risk of delinquency or default IDR may operate less like an implicit tax and more like insurance: by eliminating fixed payment obligations that cannot be met during unemployment spells, it reduces the cost of job search and allows borrowers to wait longer for better matches. If this insurance channel dominates, IDR expansion could actually *raise* employment rather than reduce it, with effects concentrated among borrowers most exposed to non-payment risk. This paper tests these predictions and quantifies the labor supply effects of IDR expansion in the US.

I use micro-data linking federal loan records to longitudinal employment histories from two nationally representative longitudinal studies from the National Center for Education Statistics (NCES): the 2008/18 Baccalaureate and Beyond Longitudinal Study (B&B) and the 2012/17 Beginning Postsecondary Students Longitudinal Study (BPS). Both datasets include matched administrative records from the National Student Loan Data System (NSLDS), enabling precise identification of borrowers eligible for the 2009 and 2015 IDR expansions alongside rich quarterly employment histories. Using Difference-in-Differences (DiD) and event-study designs, I compare labor market outcomes for borrowers exposed to each expansion against similar peers without eligible federal loans.

I present three main findings. First, and most central, the introduction of Income-Based Repayment (IBR) in 2009 significantly increased the probability of employment for exposed bache-

lor's degree recipients by 2.1 percentage points (2.6 percent) relative to comparison peers. This effect persisted for at least two years post-expansion, reflecting a durable change in labor force attachment rather than a transitory response. The employment gains were concentrated precisely where the insurance channel predicts among lower-achieving, at-risk borrowers most exposed to non-payment risk. Borrowers in the bottom tercile of standardized test scores were 4.2 percentage points more likely to be employed following the expansion, compared to an imprecise 1.2 percentage point increase among top-tercile achievers. Second, I find no evidence of moral-hazard reductions in labor supply on the intensive margin. Average hours worked and hourly wages did not decline following either the 2009 or 2015 expansions, consistent with the US design that lacks the salient income thresholds that generated bunching in the Australian setting (de Silva, 2025). Furthermore, this aggregate null result masks racial heterogeneity where Black borrowers experienced hourly wage gains of 5–6 percent following both expansions. This finding is consistent across both expansions but the mechanism warrants further investigation. Third, distributional analysis of monthly earnings confirms the 2009 expansion shifted individuals out of zero earnings and into the middle of the earnings distribution ($3,000–$5,000 per month in 2019 dollars), consistent with transitions into stable employment rather than marginal work.

This paper contributes new evidence on the labor market effects of IDR in the United States. The existing domestic literature has linked IDR primarily to repayment outcomes including reductions in the probability of delinquency (Herbst, 2023), default (Mueller & Yannelis, 2019a), and temporarily lower loan balances (Herbst, 2023). These results establish IDR as a tool for smoothing consumption and avoiding negative repayment events, but say little about labor supply. The closest domestic exception is Murto (2023), who links IDR expansion to field-of-study choices and earnings trajectories; I complement that work by focusing on employment and hours among borrowers who had already made those choices. Outside the United States, income-contingent loan policies have been linked to small labor supply reductions (intensive margin) in the United Kingdom (Britton & Gruber, 2020) and Australia (Chapman & Leigh, 2009; de Silva, 2025). My findings are not in tension with this evidence; the contrasting results across settings suggest that design features like the presence or absence of salient repayment thresholds can help pin down whether moral-hazard or an insurance channel will prevail.

I also contribute to a broader literature on student loan debt and labor market decisions. When debt obligations are removed or reduced, borrowers tend to take lower-paying jobs with higher amenity value (Rothstein & Rouse, 2011; Field, 2009; Hampole, 2022), a substitution driven by

the reduced urgency of meeting fixed payments (Luo & Mongey, 2019). Conversely, high debt burdens can accelerate job acceptance of lower-quality matches (Ji, 2021; Weidner, 2016). Evidence from willingness-to-pay experiments confirms that IDR framing shifts preferences toward riskier, higher-expected-earnings career paths (Abraham et al., 2018). My results are consistent with this mechanism operating on the acceptance margin in a slack labor market. IDR reduces the cost of extended job search and shifts borrowers into more stable employment.

Finally, this paper contributes to a broader literature on social insurance and labor supply. Unemployment insurance has long been linked to unemployment duration and job-seeking behavior through reservation wage and risk-aversion channels (Acemoglu & Shimer, 1999; Chetty, 2008; Mortensen, 1977). A broader set of programs, including the EITC (Eissa & Liebman, 1996; Meyer & Rosenbaum, 2001), bankruptcy protection (Dobbie & Song, 2015), parental co-residence (Kaplan, 2012), and public health insurance (Garthwaite et al., 2014), similarly shape labor supply by altering the financial consequences of employment transitions. IDR shares the consumption-smoothing objective of these programs without providing a direct transfer, and my results suggest it operates through a similar channel. Ganong et al. (2024) show that even very large UI supplements had small effects on job-finding relative to their consumption-smoothing effects, suggesting that when program design lacks salient earnings penalties, insurance value tends to dominate labor supply distortions, consistent with the pattern documented here.

The findings carry forward-looking implications as well. The US income-driven repayment landscape has changed substantially since the expansions studied here. Recent legislation has consolidated and restructured IDR options, reducing the generosity of payment terms available to lower-income borrowers and curtailing the zero-payment floor that previously shielded unemployed borrowers from fixed obligations. If the employment gains documented here operated primarily through the insurance channel—lowering the cost of job search for borrowers most exposed to default risk—then a contraction in IDR generosity would be expected to mute precisely those gains among the most vulnerable borrowers. Whether the new repayment architecture replicates the labor market benefits of earlier expansions is an open question, and one this paper's framework is well positioned to help inform.

The paper proceeds as follows. Section 2 briefly describes the institutional history of IDR in the US. Section 3 develops a simple model of labor force participation and job search that formalizes the insurance channel predictions and contrasts them with the intensive-margin moral hazard predictions. Sections 4–5 describe data and identification. Sections 6–7 present results for the 2009 and

2015 expansions and a series of robustness checks supporting the main results. Section 8 concludes with policy implications and directions for future research.

2. Background

Federal student loans are repaid under one of two broad structures. Under a *standard fixed plan*, monthly payments are determined at origination based on the outstanding balance and a fixed interest rate, with full repayment expected within ten years. Payments are constant and do not adjust to fluctuations in the borrower's income; failure to pay triggers delinquency, and sustained non-payment leads to default, with consequences including credit damage, wage garnishment, and seizure of tax refunds.

IDR plans offer an alternative in which monthly payments are tied to earnings rather than balance. Under most IDR plans, required payments are a fraction of discretionary income, defined as adjusted gross income above a multiple of the federal poverty guideline, and any remaining balance is forgiven after a set number of years. For borrowers earning below the income allowance, required payments are zero, eliminating the threat of delinquency during periods of low earnings or unemployment. Table 1 summarizes the key features of US IDR plans since their introduction.

The first IDR plan, Income-Contingent Repayment (ICR), was introduced in 1994 but remained little-used for over a decade, primarily because its payment rate of 20 percent of discretionary income offered modest relief over the standard plan and eligibility was limited to Direct loan borrowers.[1] The landscape changed materially in 2009, when the Obama administration introduced Income-Based Repayment (IBR)—the first expansion I study.

2.1. The 2009 IBR Expansion

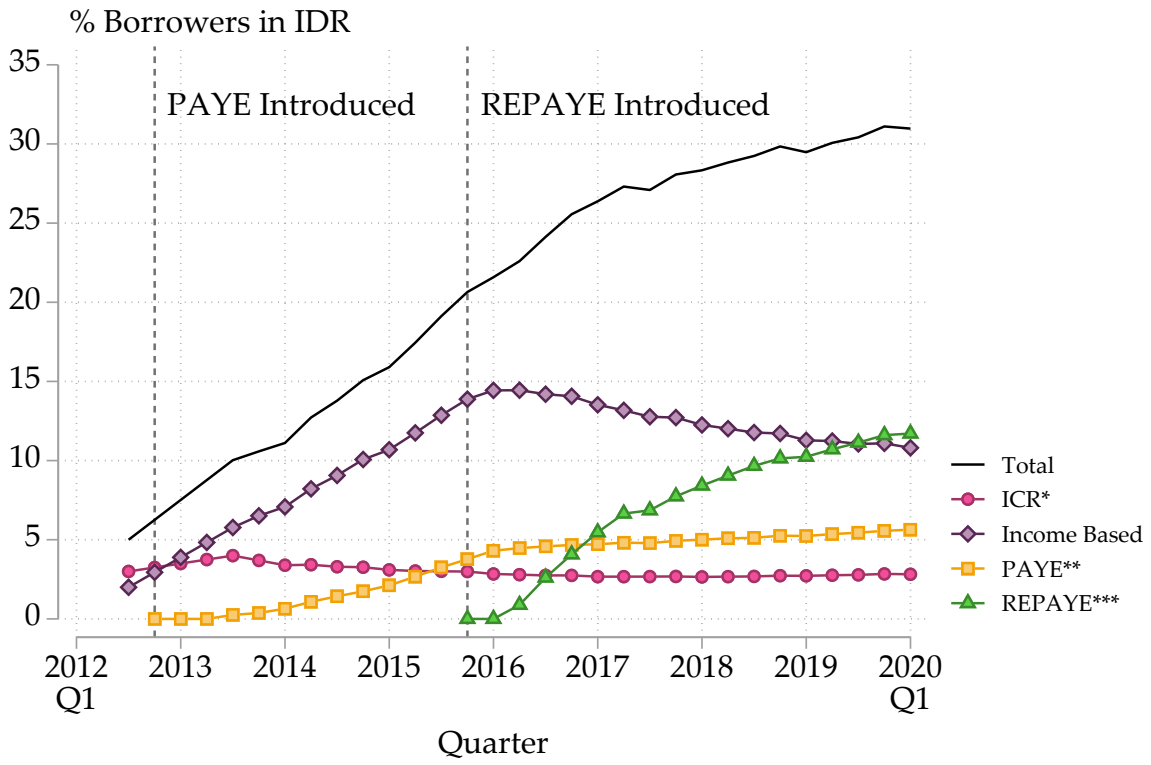
IBR represented the largest expansion of income-driven repayment to that point. It reduced the payment rate to 15 percent of discretionary income (from 20 percent under ICR), extended eligibility to both Direct and FFEL borrowers, and set a forgiveness horizon of 25 years. Crucially, IBR's income allowance protected earnings up to 150 percent of the federal poverty guideline, meaning borrowers earning near the poverty line owed nothing. The plan was finalized and made available to borrowers in July 2009—the trough of the Great Recession—when unemployment among recent college graduates was elevated and default risk was high.

[1] In that era, a substantial share of borrowers held debt originated under the Federal Family Education Loan (FFEL)) program, which was funded by private banks but guaranteed by the federal government. The FFEL program was abolished in 2010 in favor of an all-Direct program, though FFEL loans remain relevant for identifying treatment in this paper.

Simultaneously, the federal government established the Public Service Loan Forgiveness (PSLF) program, which promises full forgiveness of remaining balances after ten years of qualifying payments under an IDR plan for borrowers employed in public service. The coincidence of IBR and PSLF creates a potential confound for the employment effects I estimate; I address this directly in Section 7.

Take-up of IBR was initially low. Figure 1 shows that fewer than 5 percent of borrowers were enrolled in any IDR plan prior to 2012, reflecting documented barriers including administrative complexity, lack of awareness, and the active opt-in requirement (Cox et al., 2020; Anderson et al., 2018). This low take-up is consequential for interpreting the employment effects I estimate: the insurance value of IDR access need not require enrollment to affect behavior, a point I return to in the framework and robustness sections.

Figure 1: Percent of federal student loan borrowers enrolled in Income-Driven Repayment Plans, 2012-2020



2.2. The 2015 REPAYE Expansion

Citing continued difficulties in accessing IDR, the Obama administration issued a directive in 2014 requiring the Department of Education to expand access further (Obama, 2014). The resulting plan, Revised Pay as Your Earn (REPAYE), was made available in December 2015 and represents the second expansion I study. REPAYE's primary innovation was eligibility: it removed the income and loan-type requirements that had limited access to earlier plans, making all Direct loan borrowers eligible for IDR regardless of income or when they borrowed. The payment rate remained at 10 percent of discretionary income, a rate already available to newer borrowers through the Pay as You Earn (PAYE) plan introduced in 2012, and the income allowance remained at 150 percent of the poverty guideline.

The 2015 expansion was therefore weaker in design than 2009 in two respects. First, the reduction in required payments was smaller for most eligible borrowers, since the 10 percent rate under REPAYE was already accessible via PAYE for those who qualified. Second, the expansion occurred in a stronger macroeconomic environment, when default risk among recent graduates was lower. Both features are relevant to interpreting the contrast between the 2009 and 2015 estimates.

2.3. Recent Policy Changes

IDR enrollment grew rapidly following REPAYE's introduction, reaching approximately one-third of all borrowers by 2020 (Figure 1). Subsequent administrations extended the program further: the Biden administration's Saving on a Valuable Education (SAVE) plan, introduced in 2023, raised the income allowance to 225 percent of the poverty guideline and reduced the undergraduate payment rate to 5 percent, substantially expanding the zero-payment floor for lower-income borrowers. SAVE was enjoined by federal courts in 2024 and eliminated in 2025.

The One Big Beautiful Bill Act (2025) restructured the repayment landscape substantially. Beginning July 1, 2026, new borrowers will have access only to a tiered standard plan and a new income-driven option, the Repayment Assistance Plan (RAP). Unlike prior IDR plans, RAP ties payments to total adjusted gross income rather than discretionary income, with a mandatory minimum payment of $10 per month that eliminates the zero-payment floor previously available to the lowest-income borrowers. Existing IDR plans, including PAYE and ICR, will be phased out by July 2028; IBR will remain available to eligible existing borrowers. These changes are summarized in Table 1 and bear directly on the forward-looking implications of this paper's findings, which I discuss in the conclusion.

Table 1: Key Features of Income-Driven Repayment Plans in the United States

Expansion	**Income-Contingent Repayment (ICR)**	**Income-Based Repayment (IBR)**	**Pay as You Earn (PAYE)**	**Revised Pay as You Earn (REPAYE)**	**Saving on a Valuable Education (SAVE)†**	**Repayment Assistance Plan (RAP)‡**
Year Effective	1994	2009	2012	2015	2023	2026
Loans Eligible	Direct only	Direct and FFEL	Direct only	Direct only	Direct only	Direct only
Payments can exceed 10-year fixed	Yes	No	No	Yes	No	Yes
Poverty Level Threshold	150%	150%	150%	150%	225%	None‡
Percent of Discretionary Income (ρ)	20	15	10	10	5 – UG, 10 – GR	1–10% of AGI‡
Forgiveness Period	25 years	25 years	20 years	20 years – UG 25 years – GR	10 years <$12k 20 years – UG 25 years – GR	30 years

Notes: Plans are listed by year of initial availability. ρ denotes the share of discretionary income owed under each plan, where discretionary income is defined as adjusted gross income less a multiple of the federal poverty guideline. †SAVE was enjoined by federal courts in 2024 and eliminated via court settlement in December 2025. ICR and PAYE were eliminated by the One Big Beautiful Bill Act (2025); existing enrollees must transition to IBR or RAP by July 2028. ‡RAP bases payments on total adjusted gross income rather than discretionary income, using a progressive schedule: 1% of AGI for $10,000–$20,000, increasing to 10% for AGI above $100,000, with a mandatory minimum payment of $10/month. Unlike prior IDR plans, RAP eliminates the zero-payment floor for the lowest-income borrowers. Available to new borrowers beginning July 1, 2026. UG = Undergraduate; GR = Graduate.

3. Conceptual Framework

I outline why IDR expansion can raise employment on the extensive margin even though standard moral-hazard reasoning points to possible reductions in labor supply on the intensive margin. The model features two labor-market states, a participation decision, and a repayment rule that varies with the policy regime. It delivers two testable predictions: (1) IDR raises the probability of employment, with effects concentrated among borrowers most exposed to non-payment risk under the fixed plan; and (2) new employment under IDR reflects better matches—the removal of the delinquency penalty allows borrowers to wait longer and accept better realizations from the wage distribution rather than accepting the first available offer under financial pressure. Both effects are amplified when the expected delinquency penalty is large, a condition most plausible in slack labor markets where extended non-employment makes default more likely.

3.1. Environment

Borrower. A risk-averse borrower has outstanding federal student loan debt and must choose whether to participate in the labor market. Non-labor income is $y_0 \geq 0$, representing household resources available regardless of employment status (savings, spousal income, family support). Per-period utility is $u(c)$ with $u' > 0$ and $u'' < 0$.

Wages and ability. Conditional on participating, the borrower draws wage offers from a distribution $F(w \mid a)$, where $a \geq 0$ indexes ability. Ability shifts F, where higher-ability borrowers tend to receive better wage offers. The borrower can improve expected match quality by remaining in the search state longer, at a cost described below.

Repayment regimes. I index the repayment regime by $\tau \in \{0, 1\}$, where $\tau = 0$ denotes the fixed plan and $\tau = 1$ denotes IDR. Under the *fixed plan*, a constant monthly payment $P_F > 0$ is due regardless of employment status. Failure to pay in any period triggers a non-payment penalty with per-period utility cost $\Phi(t_U)$, where $t_U \geq 0$ denotes the number of periods of non-payment. I assume $\Phi' > 0$ and $\Phi'' > 0$: the penalty is increasing and convex in duration, reflecting the escalating consequences of delinquency through credit score damage in early periods and wage garnishment and formal default in later ones.

Under the *IDR plan*, the required payment depends on earnings $y \equiv wh$:

$$P_I(y) \;=\; \rho\,(y - \kappa)_+, \tag{1}$$

where $\kappa > 0$ is the protected income allowance (a multiple of the federal poverty guideline) and $\rho \in (0,1)$ is the share of discretionary income owed. When earnings fall below κ, no payment is required and no delinquency penalty applies. When the borrower is unemployed ($y = 0$), the required IDR payment is zero by construction.

Search cost and match quality. Participating in the labor market requires the borrower to remain in a search state for at least one period at direct cost $c > 0$ (time, effort, foregone leisure). Longer search produces better expected matches: let $\overline{w}(t)$ denote the expected wage from t periods of search, with $\overline{w}' > 0$ and $\overline{w}'' < 0$. Search yields improving but diminishing returns in match quality. The direct search cost $c(t)$ is identical across regimes and IDR does not change the time or effort required to find a job.

Consumption. Let $P_\tau(y)$ denote the expected payment due under regime τ at earnings y, and $\Phi_\tau(t_U)$ the associated delinquency cost, where the expectation is taken over the probability of IDR enrollment:

$$P_\tau(y) \;=\; (1-\tau)\, P_F \;+\; \tau\, \mathbb{E}[P_I(y)], \qquad \Phi_\tau(t_U) \;=\; (1-\tau)\, \Phi(t_U). \tag{2}$$

Consumption in the search state and in employment are, respectively:

$$c_S^\tau \;=\; y_0 \;-\; \Phi_\tau(t_U), \tag{3}$$

$$c_E^\tau(y) \;=\; y_0 \;+\; y \;-\; P_\tau(y). \tag{4}$$

Note that the search cost c enters the participation value function directly as a utility cost and does not reduce consumption in the search state. Under the fixed plan, each period of non-employment erodes consumption through the accumulating delinquency cost $\Phi(t_U)$. Under IDR, search-state consumption is simply y_0, independent of how long the borrower has been out of employment.

3.2. The Participation Decision

A borrower with outstanding student debt decides whether to enter the labor market or remain out of the labor force. A non-participant consumes y_0 each period with no search cost and no delinquency exposure. A participant incurs search cost c and, under the fixed plan, faces the delinquency cost $\Phi_0(t_U)$ for each period in the search state.

The value of participation under regime τ is:

$$V_P^\tau \;=\; \mathbb{E}[u(c_E^\tau(y))] - c, \tag{5}$$

where the expectation is taken over the wage distribution $F(w \mid a)$ and the probability of IDR enrollment. The value of non-participation is:

$$V_{NP} \;=\; u(y_0). \tag{6}$$

The borrower participates whenever $V_P^\tau \geq V_{NP}$.

Effect of IDR on participation. Under the fixed plan, delinquency costs accrue equally whether the borrower is searching or out of the labor force. Under IDR, delinquency costs are zero in both states. In either case delinquency costs cancel from the participation comparison. Search costs are also identical across regimes by assumption. The participation decision thus turns entirely on expected employment consumption, and specifically on the expected payment differential between regimes. For low-ability borrowers whose expected earnings fall below the allowance κ, the expected IDR payment is zero or near zero, whereas the fixed plan requires P_F regardless of earnings. The gain from switching to IDR on the participation margin is therefore:

$$\mathbb{E}\left[c_E^1(y)\right] - \mathbb{E}\left[c_E^0(y)\right] \;=\; \mathbb{E}[P_F - P_\tau(y)] \;>\; 0, \qquad y \leq \kappa, \tag{7}$$

which by $u' > 0$ implies $V_P^1 > V_P^0$ for these borrowers. Under the fixed plan some found participation unattractive because low expected consumption in employment made search not worth the direct cost, but under IDR the reduction in expected payments makes participation worthwhile.

Prediction 1. IDR expansion raises the probability of employment through the participation margin. The effect operates entirely through the reduction in expected payments upon employment and is largest for borrowers whose expected earnings fall near or below κ—that is, low-ability borrowers most exposed to non-payment risk under the fixed plan.

3.3. The Duration Decision

Conditional on participating, the borrower chooses how long to search before accepting a job. The wage distribution $F(w \mid a)$ and direct search cost $c(t)$ are identical across regimes. IDR changes

neither what employers offer nor the effort required to find them. The only force that differs across regimes is the delinquency cost $\Phi_\tau(t_U)$, which under the fixed plan accumulates with each additional period of search and under IDR is identically zero.

Each additional period of search yields a better expected match, because $\overline{w}(t)$ is increasing in t, but at a cost. Under the fixed plan, the marginal cost of one more period of search has two components: the direct search cost $c'(t)$ and the marginal delinquency cost $\Phi'(t_U)$. Under IDR, only the direct search cost applies. Let $t^*(\tau)$ denote the optimal search duration under each repayment regime. The borrower's optimal stopping condition is:

$$\overline{w}'(t) \;=\; c'(t) \;+\; \Phi'_\tau(t_U), \tag{8}$$

where $\Phi'_0(t_U) > 0$ under the fixed plan and $\Phi'_1(t_U) = 0$ under IDR.[2] Because the right side is strictly larger under the fixed plan, the fixed-plan borrower stops searching earlier:

$$t^*(1) \;>\; t^*(0). \tag{9}$$

Intuitively, the delinquency clock creates urgency because each additional period of search under the fixed plan is costly in time and increasingly costly in delinquency consequences. IDR removes this urgency entirely, allowing the borrower to wait longer, reject low draws, and hold out for better realizations from the same wage distribution $F(w \mid a)$.

Prediction 2. IDR expansion improves the quality of employment matches among newly employed borrowers. The effect operates entirely through the removal of the delinquency cost as a marginal cost of continued search and is independent of the payment reduction that drives Prediction 1.

3.4. Heterogeneity by Ability and Default Risk

Both predictions are amplified for low-ability borrowers, but through distinct channels. In the participation decision, the expected payment differential $\mathbb{E}[P_F - P_\tau(y)]$ is largest for borrowers whose expected earnings fall near or below the allowance κ. High-ability borrowers expect to earn well above κ, so expected IDR payments approach P_F and the participation gain shrinks. In the duration decision, the delinquency cost $\Phi'(t_U)$ is most binding for borrowers who require long

[2] Since $\overline{w}'' < 0$ and $c'' > 0$, the left side is decreasing and the right side is increasing in t; a unique optimal search duration, $t^*(\tau)$ for each regime is well-defined.

search spells to find acceptable matches. High-ability borrowers receive good offers quickly and are therefore less affected by the removal of the delinquency cost as a search friction. The model thus predicts that both the participation effect and the match quality effect are concentrated among low-ability, high default risk borrowers.

3.5. Relation to the Intensive Margin

A richer search model with endogenous hours would predict a reduction in hours for workers earning above the income allowance κ as a standard implicit-tax effect of income-contingent repayment. Whether this distortion is detectable depends critically on whether the repayment schedule creates a salient notch under which to bunch earnings. de Silva (2025) finds precisely this in the Australian system, where sharp income thresholds generate bunching just below repayment cutoffs. The US IDR system has no equivalent notch. Payments taper smoothly above κ, the marginal rate ρ is low, and enrollment requires active opt-in. The implicit-tax distortion is present in principle but diffuse in practice, making the extensive-margin insurance channel the dominant prediction in the US setting. The two sets of findings are complementary, as the Australian system's sharp income thresholds make the intensive-margin distortion dominant, while the US system's smooth phase-in and active opt-in requirement make the extensive-margin insurance channel more prominent.

4. Data and Sample Construction

To test these theoretical predictions I use two nationally representative longitudinal studies of college students developed and distributed by NCES—the B&B and the BPS. In the following sections I describe the construction of longitudinal measures of employment and student loan information along with the identification of suitable treatment and comparison groups for each of the IDR expansions using these two restricted-use datasets.

4.1. Baccalaureate and Beyond Longitudinal Study

B&B is a nationally representative sample of students who completed bachelor's degrees in the 2007–08 academic year (July 1, 2007 - June 30, 2008). Initially sampled and interviewed as part of the 2008 National Postsecondary Student Aid Study (NPSAS:08) this B&B cohort was subsequently interviewed again in 2009, 2012, and a final time in 2018. Each round of interviews collected information on employment, earnings, job amenities, and satisfaction among several other topics including post-baccalaureate education, family formation, and household finance. The data

also include matched administrative records from the NSLDS, a database that tracks federally distributed financial aid, namely Pell Grants and student loans. In the case of loans, the repayment histories and administrative events like default, deferment, and repayment plans are also tracked and included with the B&B data.

In order to evaluate employment outcomes longitudinally, I reshape the data into quarterly histories, focusing on variables related to employment, student loan repayment, post-baccalaureate enrollment (e.g., graduate school), and changes to family size from marital or childbirth events.

Employment and earnings histories were collected for each employer for which a respondent reported working. Respondents were asked about their tenure dates, starting and ending wages, and average hours worked per week at each job. I combined information from the 2012 and 2018 interviews, which together provide full 10-year employment histories beginning with the quarter after bachelor's degree completion through the quarter during which the final interview was taken in 2019. The maximum number of employers reported in the early period between 2008 and 2012 was 12 and between 2012 and 2018 the maximum was 14, with a mean of 2 and 3.2 respectively. I adjust all earnings for inflation using quarterly values from the U.S. Bureau of Economic Analysis Personal Consumption Expenditure (PCE) seasonally adjusted index, retrieved online from the Federal Reserve Bank of St. Louis. All dollar estimates in this paper are reported in 2019 (quarter 1) dollars unless noted otherwise.

Data on student loan borrowing and repayment were taken from NSLDS source files tracking disbursements, outstanding balance and interest, repayment plans, deferment, default, and delinquency events for each loan. To identify treated borrowers, I start by excluding all Parent PLUS loans, as the obligation for repayment does not rest with the B&B sample member, but with their parents instead. Loans tracked in NSLDS include both those made through the Direct Loan and FFEL programs. To most borrowers, this difference is immaterial, however, the administrative difference creates complications in making new policy, particularly with programs like IDR, where some loan types are deemed ineligible for enrollment. A benefit of the data used here is the inclusion of both types, which is not always possible (e.g., Herbst, 2023).

To be eligible for the 2009 IBR expansion sub-sample, I require individuals to have separated from school for at least 4 quarters (1 year) prior to the July 2009 introduction date. This timing makes a majority of the sample eligible for this expansion since a requirement to be in B&B is to have finished a bachelor's degree in the 2007-08 academic year. Thus, most students finished their bachelor's degrees in the 2nd quarter of 2008 and the second most common exit was in quarter 4

of 2007. This excludes individuals who re-entered some form of higher education (e.g,. graduate school) immediately after their bachelor's degree.

I define treated individuals as those who had federal student loans in repayment prior to the expansion. A small contingent of borrowers pay off their loans almost immediately following the completion of their bachelor's degree and prior to the 2009 expansion. I consider these borrowers untreated, and instead as part of the comparison group. IDR should have no effect on their labor market choices, as they had no loans subject to the new policy. The majority of the comparison group are non-borrowers, or those who only borrowed private student loans for college. Of the 9,350 students in the full sub-sample, about 63 percent were in repayment on federal student loans and part of the "treatment" group. Additional summary statistics for the sample can be found in the first two columns of Table 2. The sample is predominantly white (74 percent), and highly likely to have participated in the labor force at least one quarter in those prior to expansion (91 percent). Over half of the sample received a Pell Grant while an undergraduate (52 percent) and over half had at least one parent with a bachelor's degree or higher (51 percent). One nice feature of IDR is that predicted monthly payments can be calculated for both borrowers and non-borrowers alike because they are a function of income and independent of any amount borrowed. It is highly co-linear with earned income.[3] In looking across this sub-sample, just over one-quarter (26 percent) had pre-period income and family sizes that qualified them for $0 payments under the new IBR plan.

4.2. Beginning Postsecondary Students Longitudinal Study

To examine the IDR expansion in 2015, I turn to the BPS, which contains a nationally representative sample of students who began college in the 2011-12 academic year (July 1, 2011 - June 30, 2012). Unlike B&B, BPS contains students from much more diverse programs and institutions, including a heavy contingency pursuing shorter-term credentials like Associate's degrees and certificates and attending two-year public or for-profit colleges and universities. BPS sample members were initially selected and interviewed for the 2012 National Postsecondary Student Aid Study (NPSAS:12) and subsequently interviewed again in 2014 and 2017, 3 and 6 years after *starting* college respectively.

I follow a similar data management procedure for creating a quarterly history of employment, enrollment, and student loan histories for each sample member as in B&B, taking into account mi-

[3]The only reason it is not is IDR payments are also a function of household size and have a floor of zero, disallowing negative values.

nor differences in interview data collection and administrative record matching. Data sourced from NSLDS is nearly identical, with one important distinction. Because this group of students started college in 2012, there exist no valid FFEL loans dispersed, because the program was discontinued in 2010.

To qualify for this analytic sample, the individual must have exited school for at least 2 quarters prior to the expansion, and to be treated, I again use borrowers with at least some federal loans in active repayment prior to December 2015. I exclude those who completed bachelor's degrees from the sample for two reasons. The first is that few had the time to complete and exit school early enough to be eligible for the expansion at the end of 2015. The second is that the experience of bachelor's degree recipients is more thoroughly covered in the B&B samples. The resulting BPS sample is similar in size to the 2009 expansion, at 9,320, but with major differences in characteristics. The relatively short time window and selection criteria makes this sample especially over-representative of stop-outs from initial college enrollment. Descriptive statistics in the last two columns of Table 2 show about two-thirds of the sample had not completed any degree or credential prior to expansion. The sample is also more racially and ethnically diverse than the B&B at 52 percent white, 18 percent Black, and 22 percent Hispanic. Almost three-quarters of this group recieved a Pell Grant and less than 10 percent had parents who obtained at least a bachelor's degree. This sample was considerably less like to have worked any quarters prior to expansion (70 percent) and over half (57 percent) had income and family sizes that would qualify them for zero dollar IDR payments under the new plan, REPAYE.

4.3. Defining treated borrowers

In general, borrowers are classified as part of the treatment group if they had active federal loans in repayment at the time a new IDR plan became available, regardless of whether their income or current monthly payments made them *eligible* for enrollment and regardless of whether or not they actually *enrolled* in the new IDR plan. I take this approach because I am interested in the effects of the policy change itself. In addition, the indirect effect or option-value of IDR could be particularly important in this case. Conditional on being aware of IDR, borrowers ineligible for the plans at the time of expansion may still make adjustments to their labor force participation in response. From the perspective of evaluating federal policy, it is useful to know if labor supply responses were observed among borrowers when repayment options were added. Selecting a treated sample by conditioning on eligibility or actual enrollment hampers this objective by

Table 2: Descriptive Statistics, by Cohort

Variable	**2009 Expansion (B&B:08/18 Cohort)**		**2015 Expansion (BPS:12/17 Cohort)**	
	Mean	SD	Mean	SD
Treated	0.630	0.483	0.603	0.489
Race/Ethnicity: White	0.736	0.441	0.520	0.500
Race/Ethnicity: Black	0.088	0.284	0.183	0.387
Race/Ethnicity: Hispanic	0.088	0.283	0.217	0.412
Race/Ethnicity: Asian	0.054	0.227	0.022	0.146
Race/Ethnicity: Other	0.033	0.178	0.057	0.233
Sex: Female	0.586	0.493	0.563	0.496
Age	25.167	6.442	22.826	7.808
Completed: No degree	-	-	0.669	0.471
Completed: Certificate	-	-	0.187	0.390
Completed: Associate's	-	-	0.144	0.351
Completed: Bachelor's	0.999	0.037	-	-
Completed: Master's	0.001	0.037	-	-
Has standardized test score	0.824	0.381	0.588	0.492
Pell Grant Recipient	0.516	0.500	0.737	0.440
Parents obtained BA or higher	0.510	0.500	0.079	0.269
Institution: Public	0.583	0.493	0.507	0.500
Institution: Private nonprofit	0.366	0.482	0.090	0.286
Institution: Private for-profit	0.051	0.220	0.404	0.491
Married	0.280	0.449	0.262	0.440
Number of Children	0.313	0.785	0.594	1.025
Pre-period monthly income (incl spouse)	3116.93	2929.47	1923.93	2410.03
IDR income level = 0	0.264	0.441	0.569	0.495
IDR income level > 0, below median	0.368	0.482	0.215	0.411
IDR income level > 0, above median	0.368	0.482	0.216	0.412
Participant in labor force (pre-expansion)	0.908	0.289	0.698	0.459
Participant in labor force (post-expansion)	0.924	0.265	0.772	0.420
Hourly Wage (pre-expansion)*	17.68	14.43	12.55	11.09
Hourly Wage (post-expansion)*	19.59	14.80	13.49	12.98
Average Hours Worked (pre-expansion)*	35.41	12.81	31.37	15.58
Average Hours Worked (post-expansion)*	37.07	13.04	33.00	14.73
Number of Individuals	9350		9320	

Notes: *Indicates only individuals with positive values were included. Pre-period monthly income is expressed in dollars unadjusted for inflation to facilitate calculation of predicted IDR payments. Hourly wage rates are expressed in 2019 dollars. Age in the B&B sample refers to age at the time of bachelor's degree. Age in BPS refers to age when beginning school.

implicitly assuming large swaths of borrowers were unaffected by the change.

Still, there are benefits to considering the use of alternative definitions of treatment and comparison groups. In section 7 I present results limiting the sample to borrowers and consider the role of direct effects of IDR enrollment as well, both of which I discuss in conjunction with the presentation of the results.

4.4. Inverse-Probability Weighting

In my preferred samples, non-borrowers make up the majority of the comparison groups. I take this approach because non-borrowers arguably have little reason to alter their short-term labor market decisions and career trajectories based on changes to student loan policy.[4] It is potentially concerning that the observable characteristics of borrowers are often different than those of non-borrowers along dimensions like initial labor market choices that could also be related to career trajectories (e.g., Hampole, 2022). In Table 3, I compare the means of treated and comparison groups across sub-samples and along selected covariates known to be related to student loan repayment and labor market outcomes. In both samples, borrowers are disproportionately non-white, from lower socioeconomic backgrounds, and have lower initial pre-period incomes.

To address concerns that these differences might yield differential labor force participation rates and earnings, I implement an inverse-probability weight (IPW) approach to balance the treated and comparison individuals along observable characteristics and pre-period outcomes. I model the probability of being in the treatment group as a function of background characteristics, pre-period household composition, and pre-period income and labor market participation. I construct weights for the full sample and separately for workers observed on the intensive margin.[5]

Weighted descriptive statistics between treated and comparison groups can be found in the last columns of Table 3. Looking at the pre- versus post-weighting standardized differences, the IPW procedure significantly reduced imbalance along the selected covariates. None of them have standardized differences greater than 0.05. In the remainder of the paper, I preference the use of these IPWs to estimate average treatment effects on the treated borrowers, though I provide unweighted estimates in some instances as well. I discuss this and further assumptions for making causal inference more in the empirical section.

[4] One alternative is to ignore non-borrowers and classify borrowers who would not be eligible for lower payments under IDR plans as the comparison group. For robustness, I take perform this exercise and discuss the results and its downsides in Section 7.

[5] Weights take the standard inverse-probability form $\text{IPW}_i = D_i/e(X_i) + (1 - D_i)/(1 - e(X_i))$, normalized to sum to one within each group following Imbens (2004), where $\mathbf{X}$ is the vector of pre-period characteristics and the treatment indicator, $D_i \in \{0, 1\}$.

Table 3: Unweighted and Inverse-Probability Weighted Means and Standardized Differences across Comparison and Treated Groups, by Sample

Variable	**Unweighted**			**Inverse-Probability Weighted**		
	Comparison	Treated	Std Difference	Comparison	Treated	Std Difference
B&B:08/18 Cohort, 2009 Expansion						
Race/Ethnicity: White	0.740	0.735	-0.011	0.726	0.731	0.011
Race/Ethnicity: Black	0.066	0.101	0.126	0.092	0.089	-0.008
Sex: Female	0.568	0.597	0.058	0.588	0.586	-0.004
Pell Grant Recipient	0.323	0.633	0.653	0.521	0.518	-0.006
Institution: Public	0.622	0.560	-0.127	0.576	0.580	0.008
Institution: Private nonprofit	0.348	0.376	0.057	0.372	0.369	-0.006
Institution: Private for-profit	0.029	0.064	0.165	0.052	0.051	-0.005
Married	0.295	0.271	-0.054	0.288	0.283	-0.012
Pre-period monthly income (incl spouse)	3301.691	3005.395	-0.101	3127.237	3131.899	0.002
Participant in labor force (pre-expansion)	0.899	0.914	0.050	0.907	0.907	-0.001
Hourly Wage (pre-expansion)*	18.48	17.22	-0.088	17.81	17.67	-0.010
Average Hours Worked (pre-expansion)*	35.69	35.25	-0.034	31.35	31.48	0.011
Number of Individuals	3520	5830				
BPS:12/17 Cohort, 2015 Expansion						
Race/Ethnicity: White	0.564	0.491	-0.148	0.524	0.534	0.020
Race/Ethnicity: Black	0.136	0.215	0.210	0.184	0.184	-0.001
Sex: Female	0.502	0.603	0.203	0.544	0.559	0.031
Pell Grant Recipient	0.586	0.837	0.575	0.735	0.727	-0.018
Institution: Public	0.751	0.346	-0.891	0.495	0.508	0.029
Institution: Private nonprofit	0.067	0.105	0.137	0.090	0.091	0.000
Institution: Private for-profit	0.183	0.550	0.824	0.415	0.401	-0.029
Married	0.278	0.251	-0.059	0.267	0.264	-0.006
Pre-period monthly income (incl spouse)	2078.87	1821.96	-0.104	1948.46	1929.31	-0.008
Participant in labor force (pre-expansion)	0.720	0.682	-0.083	0.69	0.697	0.009
Hourly Wage (pre-expansion)*	12.91	12.28	-0.056	12.69	12.62	-0.007
Average Hours Worked (pre-expansion)*	31.52	31.25	-0.017	31.48	31.63	0.009
Number of Individuals	3700	5620				

Notes: *Indicates only individuals with positive values were included. Pre-period monthly income is expressed in non-inflation adjusted dollars to facilitate calculation of predicted IDR payments. Std=Standardized, a difference calculated by taking difference in means between the two groups divided by the square root of the average of the two group variances.

5. Empirical Strategy

This setting with a binary indicator for exposure to a new policy paired with panel data lends itself to DiD and event study designs to test my empirical questions of interest. I am primarily interested in the effects of gaining access to a new IDR plan, not the effects of actually enrolling in them. This amounts to an intent-to-treat (ITT) effect because the majority of borrowers who gain access to IDR will not use them but may still respond to the insurance they provide by adjusting job search and labor supply. The general DiD equation of interest is,

$$Y_{ict(v)} = \beta_1 \left(D_i \cdot P_{ct(v)} \right) + \delta_{t(v)} + \delta_i + \epsilon_{ict(v)}, \tag{10}$$

where Y represents the outcome including a binary indicator for employment, log hourly wages, and log weekly hours worked, for each individual from expansion cohort $c \in \{2009Q3, 2015Q4\}$ and at quarter t.

The parameter of interest, β_1, comes from the interaction of the treatment indicator D_i and post-expansion indicator $P_{ct(v)}$. I control for quarter-specific time shocks across individuals, but as a function of exit cohort, v, so that comparisons are within exit group defined by the six-month period (or term) in which they left school prior to expansion (e.g., graduated). I do this to avoid comparing individuals with differing levels of work experience and to increase the likelihood of meeting the necessary parallel trends assumption described in the next paragraphs. The equation is rounded out with individual fixed effects and an idiosyncratic error term.

At its core, this specification is a Two-Way Fixed Effects (TWFE) design. However, I avoid issues now well-known and documented in the econometric literature of negative weights with these estimators (e.g., Goodman-Bacon, 2021; de Chaisemartin & D'Haultfœuille, 2020) by separating the cohorts only ever comparing individuals within the same treatment cohort and with a common treatment intervention.

In order for this method to yield credibly causal estimates I rely first on the "no-anticipation assumption."[6] In the context of IDR expansion, this requires students not to have adjusted their job choices prior to the actual availability of the new plans. Pending changes to plans were often announced by the administration ahead of time, but the details were unclear until the dates of implementation.

[6] Let $Y_{i,t}(0)$ and $Y_{i,t}(1)$ denote the potential outcomes for individuals in quarter, t. This assumption requires: $E[Y_t(1) - Y_t(0)|D = 1] = 0$, $\forall t < c$.

The second assumption requires the outcomes in the untreated state between treatment and comparison groups to evolve in parallel, conditional on covariates, in my case the fixed effects included in Equation 10.[7]

Though this assumption is technically unverifiable due to the inability to observe both potential outcomes for each group, there are several tests that can lend credibility to the assumption. Wooldridge (2021) offers two such options. The first, is an event-study type linear regression of the outcome onto the treatment indicator D_i interacted with quarter indicators in all the pre-expansion quarters,

$$Y_{ict(v)} = \sum_{\tau=1}^{c-2} \beta_\tau \left(D_i \cdot 1\{t = \tau\} \right) + \delta_{t(v)} + \delta_i + \nu_{ict(v)}, \ \forall t < c \tag{11}$$

Post-estimation, I run a joint significance test on all the indicators β_τ. A rejection of the null that these are jointly equal to zero may suggest a violation of parallel linear trends in the outcome between treated and comparison units. One critique of pre-testing of this nature is that the tests are often under-powered and we might under-reject the null of no pre-trends (Roth, 2022). One way to save degrees of freedom with a similar pre-test is to replace the quarter-time indicators in Equation 11 with a single parameter for the linear time-trend interacted with the treatment indicator. Post-estimation, I test the significance of just one coefficient, though the evaluation criteria is similar. A rejection of the null suggests the linear trends are not significantly different in the pre-periods, which is supportive evidence that the comparison unit outcomes were evolving similarly (in a linear sense) to the treated units prior to IDR expansion. These tests can also be augmented to examine trends in sub-groups of individuals, which is important in my setting when looking at heterogeneity of estimates across observable characteristics.

6. Results

The results that follow assess the two predictions from Section 3, on the probability of employment and match quality among newly employed borrowers, separated by the 2009 and 2015 expansion cohorts.

6.1. 2009 Expansion

The 2009 IDR expansion generated positive effects on the extensive margin of work. Borrowers exposed to the expansion were significantly more likely to be employed in the quarters fol-

[7] Formally, $E[Y_t(0) - Y_{t-1}(0) \mid X, D = 0] = E[Y_t(0) - Y_{t-1}(0) \mid X, D = 1]$ for all t.

lowing its introduction, with effects concentrated among those with lower academic achievement and greater pre-expansion non-payment risk. The distributional evidence is consistent with better match quality among new entrants. Changes to the intensive margin were largely null in aggregate, with the notable exception of Black borrowers, whose wages increased significantly following the expansion.

6.1.1. Main effects

Table 4 contains results for the main DiD specifications from Equation 10 for the 2009 expansion for three outcomes—the probability of employment, logged hourly wages and logged hours worked per week. I include both unweighted and IPW estimates along with the p-values from the two tests for pre-trends described in the methods section.

With respect to the pre-trend tests, I am unable to reject the null hypotheses of no pre-trends at conventional levels of confidence for any specification. Though not sufficient for a causal interpretation of the results, this helps to bolster the necessary parallel trends assumption. In general, the case for parallel trends is stronger in the IPW estimates, though differences in the p-values are marginal.

The 2009 IBR program expansion yielded a positive effect on the extensive margin. Post-expansion, borrowers were between 1.1 and 2.1 percentage points more likely to be employed compared to non-borrowers. The weighted sample, which strongly balances the groups along observable characteristics including pre-period labor force participation, is statistically significant at the 1 percent level. The baseline employment rate among this group is about 83 percent across pre-expansion quarters, equating the (weighted) treatment effect with about a 2.6 percent increase in the probability of employment.

On the intensive margin, I find null effects for hourly wage rates, consistent across both weighted and unweighted estimates. In terms of hours worked, the estimates point more toward a positive effect, at around 1 percent, but these are also imprecise.

6.1.2. Event studies

I also test for the presence of dynamic treatment effects, generating event study estimates using the following general specification,

Table 4: Difference-in-Differences Estimates on the Probability of Employment, Log Hourly Wages and Hours Worked

2009 Expansion (B&B:08/18)	**Employed (Y/N)**		**Log(Hourly Wages)**		**Log(Hours Worked)**	
	Unweighted	IPW	Unweighted	IPW	Unweighted	IPW
Post-expansion x Treated	0.011*	0.021***	-0.010	-0.008	0.008	0.010
	(0.006)	(0.007)	(0.007)	(0.008)	(0.007)	(0.007)
N individual-quarters	98350	98030	79030	78910	79450	79320
N Individuals	9390	9350	8160	8140	8170	8160
Event study p-value	0.30	0.90	0.60	0.97	0.35	0.60
Heterogeneous trend p-value	0.24	0.41	0.69	0.68	0.26	0.25

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The main estimates come from the DiD specification in Equation 10 in the main text. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort.

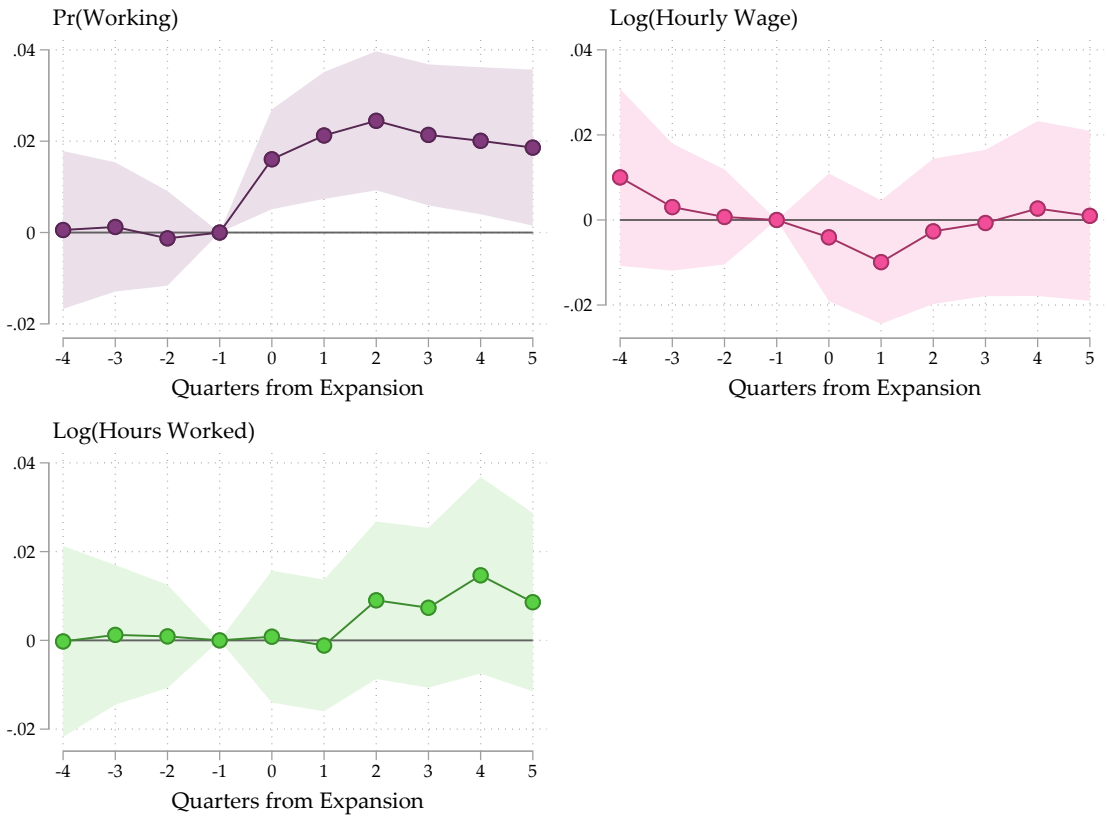
$$Y_{ict(v)} = \sum_{\tau \neq -1} \beta_\tau \left(D_i \cdot 1\{t = \tau\}\right) + \delta_{t(v)} + \delta_i + \nu_{ict(v)}, \ \forall t \in \{c-4, ..., c+5\}. \tag{12}$$

I then plot the estimates of the weighted β_τ's along with 95 percent confidence intervals each of the three outcomes in Figure 2.

The first panel in **purple** shows the dynamic effects of the probability of employment. Here, the aggregate effect size of 2.1 percentage points is evident by an immediate 2 percentage point jump in the probability of employment in the first quarter after expansion, an effect that rose slightly in the next two quarters and also remained relatively constant for all 6 quarters estimated here. This pattern is suggestive of marginal borrowers' movement into stable employment, rather than transient or short-run seasonal jobs.

In general, the intensive margin estimates are noisy. The logged hourly wage plots in **pink** in the second panel show small and insignificant differences across the period of interest. The **green** estimates for logged weekly hours worked, there is a small increase that begins 2 quarters after the 2009 expansion, but in no period do the differences rise to a conventional level of statistical significance.

Figure 2: 2009 Expansion Event Study Plots of Employment, Log Hourly Wage, and Log Hours Worked



Notes: The estimates are generated from linear regressions of the general form described in Equation 12 and are weighted by the IPW described in the main text. All figures include shaded regions for the 95 percent confidence interval of each estimate. Underlying sample comes from B&B.

6.1.3. Distributional effects

Next, I estimate the distributional effects of IDR expansion by combining the effects found on the extensive margin with those on the intensive margin. I aim to understand if the distribution of monthly earnings changed in a meaningful way due to IDR expansion and the location of any shifts within that distribution. I create 10 bins of monthly income starting with zero and in $\$1,000$ increments and assign individuals in each sample to a bin based on their average monthly earnings in all pre- and post-expansion quarters.[8] I then model the probability of appearing in each of these bins separately, in a DiD setup as follows:

$$Earn^{n}_{ict(v)} = \beta^{n}\left(D_i \cdot P_{ct(v)}\right) + \delta_{t(v)} + \delta_i + \epsilon_{ict(v)} \; \forall n \in \{\$0, \$1000, ..., \$9000\}. \tag{13}$$

Each β^n represents the change in a borrower's probability of appearing in that bin compared to the comparison group's change in the bin, pre- and post-expansion.

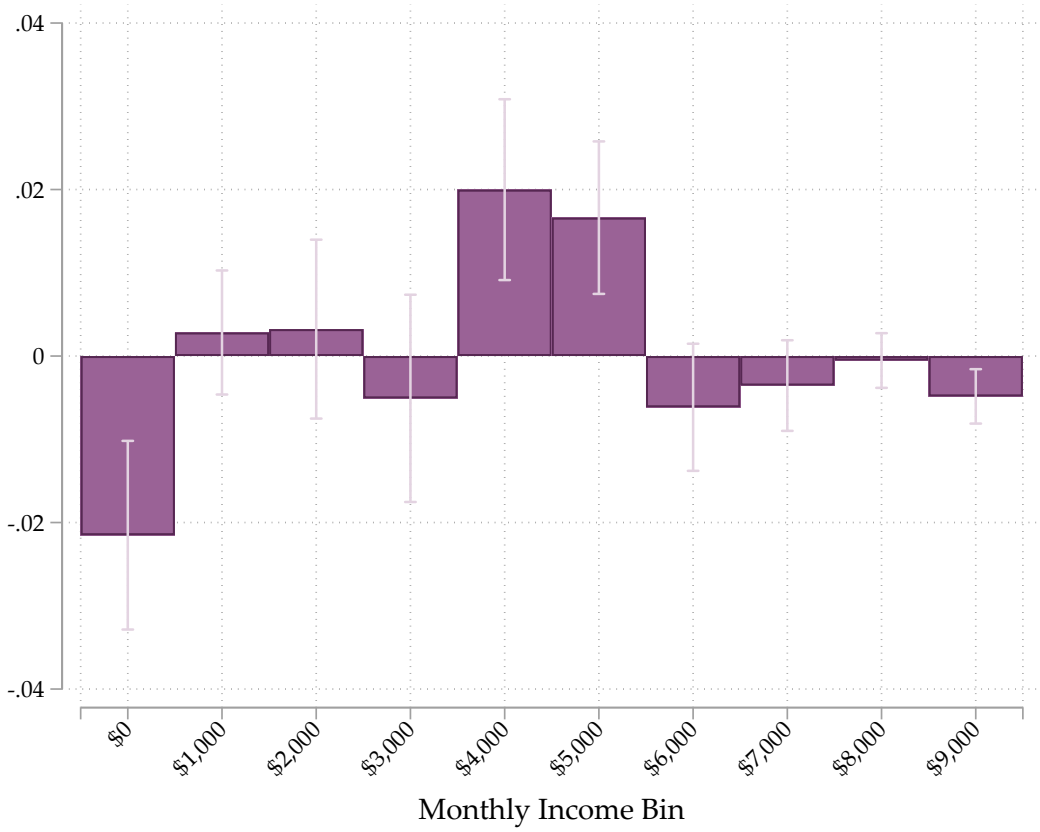
Figure 3 plots these estimates and confidence intervals for each expansion separately. The figure shows a decline in the lowest income bin on the order of around 2 percentage points. This is consistent with the positive effect from the main DiD estimate on the probability of working. The figure also shows the declines in non-workers were offset by increases in the probability of working near the middle of the income distribution, between 3 and 5 thousand dollars per month in 2019 dollars. This again supports the notion that shifts into the labor force were to more stable, likely full-time jobs, rather than marginal part-time or low-wage work. With the 2009 expansion coming during one of the largest recessions on record in the US, insurance from IDR plans appears to have encouraged job search and eventual labor force participation, with the shift toward middle earnings consistent with better match quality among new entrants.

6.1.4. Job search mechanism

Quarterly measurements distinguishing unemployed workers from workers out of the labor force (not searching for work) are not available in the B&B data. However, there are variables that capture the number of months individuals were in each of three key categories (employed, unemployed, and out of the labor force) measured after bachelor's degree completion and pre-2009 expansion and again post-expansion from 2009 through 2012. This structure allows for a canonical two-period DiD analysis. The three estimates comparing the difference in number of

[8] This binned DiD or event study approach is modeled somewhat after Cengiz et al. (2019).

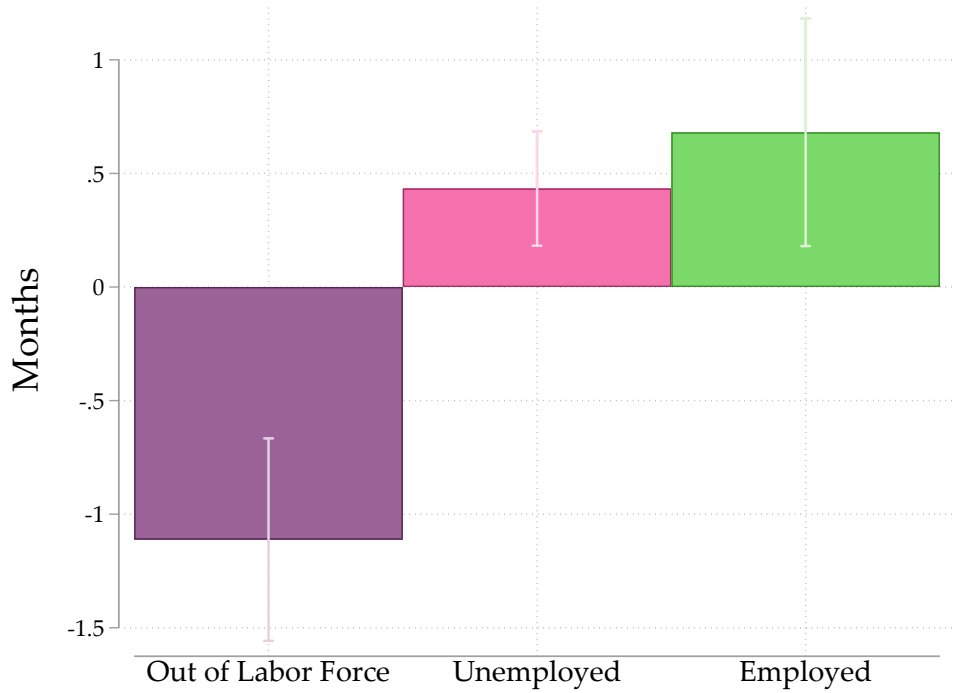
Figure 3: 2009 Expansion Effects on the Monthly Wage Distribution



Each bar and represents a DiD estimate along with a 90% confidence interval where the outcome is the probability of earning monthly wages in a given bin (e.g., $1,000) and the coefficient represents the difference pre- and post-IDR expansion between the treatment and comparison groups.

months pre- and post-expansion between treated borrowers and comparison non-borrowers will sum to zero since the categories are mutually exclusive.

Figure 4: 2009 Expansion two-period difference-in-differences estimates of months in each labor force participation status



Notes: Each bar reflects a 2-period DiD estimate of the number of months borrowers versus non-borrowers spent in one of the three labor force statuses between 2008-2009 (pre-IBR introduction) and months post-expansion from 2009-2012. The regressions control for exit cohort, pre-period labor force participation, gender, race, age and age-squared, marital status, number of children, college GPA, bachelor's institution control (public, private, or for-profit) and selectivity (high, medium, or low), and IDR income group (zero, below-median positive, and above-median positive). Standard errors are robust and the error bars depict the 90 percent confidence intervals.

Figure 4 plots the DiD estimates and their associated 90 percent confidence intervals. The pattern is consistent with Prediction 1. Borrowers exposed to the expansion spent about 1.2 fewer months "out of the labor force" compared to non-borrowers, offset by roughly 0.5 more months unemployed and 0.7 more months employed. The movement is from non-participation into active search and then employment, the participation margin the model identifies as the primary channel through which IDR raises employment.

6.1.5. Heterogeneity by achievement and non-payment risk

The model predicts that the participation effect will be largest for borrowers whose expected earnings fall near or below the income allowance κ, because the expected payment differential between the fixed plan and IDR is greatest for this group. Low academic achievement and attendance

at less selective institutions are plausible proxies for low expected earnings and, by extension, high non-payment risk under a fixed plan. I test whether these borrowers were more responsive to the expansion.

Table 5 shows that when estimating separate effects for borrowers in different terciles of the standardized test distribution[9], those in the lowest tercile were 4.2 percentage points more likely to be employed post-expansion, while those in the highest tercile have an effect indistinguishable from zero on this margin. A similar pattern emerges in comparing students receiving bachelor's degrees from low or non-selective institutions compared to highly selective institutions. Borrowers in the low selectivity category were 3.2 percentage points more likely to be employed post-expansion relative to non-borrowers attending similar institutions, versus an imprecise 1.4 percentage point difference for highly selective institution graduates. Heterogeneity estimates by achievement on the intensive margin of work did not reveal any significant differences but are available in Appendix Table A1.

Table 5: Heterogeneous effect estimates by standardized test score tercile and college selectivity, 2009 expansion

	Test Scores				**BA Institution Selectivity**		
	Lowest Tercile	Middle Tercile	Top Tercile	No Scores	Low	Medium	High
Employed (Y/N)							
Post-expansion x Treated	0.042**	0.028**	0.016	0.005	0.032**	0.019**	0.014
	(0.021)	(0.012)	(0.010)	(0.015)	(0.016)	(0.008)	(0.018)
N individual-quarters	11,910	30,050	38,390	17,680	19,790	64,310	13,930
N Individuals	1,130	2,860	3,720	1,650	1,860	6,140	1,360
Event study p-value	0.92	0.61	0.88	0.37	0.69	0.81	0.30
Heterogeneous trend p-value	0.92	0.81	0.48	0.59	0.17	0.84	0.39

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The main estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Test score terciles draw from the ACT national distribution from 2007: Low < 19, Middle $19 \leq TS \leq 23$, High > 23. Selectivity draws from the Barrons Competitiveness Index, where High selectivity includes the top 2 categories (most and highly competitive), Medium includes categories 3 and 4 (very and competitive), and Low includes all others including unclassified. All models include individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator.

As a more direct test of sensitivity to non-payment penalties I identify borrowers with delinquency or use of deferments for economic hardship prior to IDR expansion. This distinguishes borrowers who were more at-risk for non-payment and associated penalties to estimate group-

[9] I use national percentiles from the ACT where the lowest tercile is a score of 18 or below, the middle is between 19 and 23, and the highest tercile is a score above 23. Students with scores from either the SAT, ACT, or both are included in this measure, as the B&B study variable "TEACTDER" includes values from the SAT converted to the ACT scale.

specific treatment effects. I augment my main DiD equation 10 to include two levels of treatment for borrowers at risk of non-payment and those not.[10]

$$Y_{ict(v)} = \boldsymbol{\beta^R} \left(D_i \cdot P_{ct(v)} \cdot 1\{Risk_i\} \right) + \delta_{gt(v)} + \delta_i + \epsilon_{ict(v)}, \tag{14}$$

where $1\{\cdot\}$ is the indicator function and $\boldsymbol{\beta^R}$ is a vector containing two effect estimates for at-risk borrowers and one for borrowers who were not.

The risk measure is borrower-specific meaning individuals in the comparison group (non-borrowers) do not have a direct correlate. I group all individuals into three pre-period income categories taking household size into account—income at or below the 150 percent poverty threshold, income above the threshold but below the median, and those with income above the median. I then include this variable as a fixed effect interacted with quarter and exit cohort to effectively compare only individuals with similar baseline household income values.

Table 6 presents these estimates for all three labor outcomes and includes the results of an F-test of equivalence between the two group estimates. At-risk borrowers prior to expansion were over 5 percentage points more likely to be employed post-expansion versus an increase of only 1.8 percentage points for other borrowers. At-risk borrowers also experienced a 3.9 percent increase in hours worked post-expansion, where other borrowers did not adjust their hours worked. Both positive effects among at-risk borrowers are significantly larger than the estimates for non at-risk borrowers.

These patterns support the prediction that the participation effect is largest among borrowers most exposed to non-payment risk under the fixed plan.

6.1.6. Heterogeneity by Race

I also examine heterogeneous treatment effects by race. Black borrowers hold a disproportionate share of student debt and face significantly higher rates of delinquency and default (Scott-Clayton, 2018; Houle & Addo, 2022), making them a natural subgroup of interest for evaluating the insurance channel.

As shown in Table 7, employment effects do not differ significantly by race. However, hourly wages for Black borrowers increased by 6.4 percent following the 2009 expansion, significantly

[10] In a separate analysis, I removed borrowers who did not meet the criteria for the new treatment groups and generated IPW-specific estimates using only the new treatment group and the new weighted average of the comparison group. The results are similar and generally exhibit larger effect sizes than those presented here in the main text. I take the approach of including two levels of treatment to facilitate more direct comparisons between the separate groups.

Table 6: Heterogeneous effect estimates by at-risk borrower status, 2009 Expansion

	Employed (Y/N)	Log(Hourly Wages)	Log(Hours Worked)
At-risk borrower (Y/N)			
Post-expansion x Treated x No	0.018***	-0.009	0.006
	(0.007)	(0.008)	(0.007)
Post-expansion x Treated x Yes	0.052***	-0.008	0.039**
	(0.015)	(0.019)	(0.016)
Yes = No (F, p-value)	(5.92, 0.02)	(0.004, 0.95)	(4.60, 0.03)
Baseline rate (Yes)	0.12	0.09	0.09
N Observations	98030	78910	79320
N Individuals	9350	8140	8160

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The estimates come from the general specification described in Equation 14 from the main text. "At-risk" is defined as borrowers with a delinquency or an economic hardship deferment prior to the 2009 IDR expansion. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort.

larger than the estimate for non-Black borrowers $(p < 0.1)$. The mechanism behind this wage response is unclear. Results not shown indicate that Black borrowers are no more or less likely to switch jobs compared to non-Black borrowers after expansion.

Notably, labor supply responses to IDR expansion do not vary significantly by sex or marital status. Estimates along these other demographic margins can be found in Appendix Table A2.

Table 7: Heterogeneous effect estimates by race, 2009 expansion

	Employed (Y/N)		Log(Hourly Wage)		Log(Hours Worked)	
	Black	Non-Black	Black	Non-Black	Black	Non-Black
Post-expansion x Treated	0.014	0.022***	0.067*	-0.014*	-0.035	0.014*
	(0.027)	(0.007)	(0.036)	(0.008)	(0.032)	(0.007)
N individual-quarters	8690	89340	6070	72840	6110	73210
N Individuals	830	8530	640	7500	650	7510
Event study p-value	0.53	0.93	0.86	0.99	0.71	0.55
Heterogeneous trend p-value	0.67	0.27	0.51	0.54	0.99	0.18

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort.

6.2. 2015 Expansion

The 2015 IDR expansion did not elicit labor supply responses among BPS sample members. The most likely explanation is that the policy change was weak relative to 2009. The primary change was a broadening of eligibility, which matters little for borrowers who could already access a comparable plan. That the null holds even in a sample with considerably higher average default risk than B&B, including many non-completers and borrowers from two-year and for-profit institutions, points toward the weakness of the policy change rather than a lack of potential beneficiaries. One finding is consistent across both expansions: Black borrowers experienced significant hourly wage gains while non-Black borrowers did not.

6.2.1. Main effects

The main effect estimates in Table 8 are generally null. Though the share of discretionary income for payments (ρ) was 10 percent versus 15 percent in the 2009 IBR expansion, this lower value was already available to most borrowers through the PAYE option, which was introduced in 2012. In this expansionary context and with a more diverse sample there is no evidence IDR expansion had an effect on their probability of working or on the intensive margin of work. There is but one point estimate near an elasticity of 0.01, log hourly wages, and it is imprecise. Dynamic effects from event study estimates show similar null findings and are available in Appendix Figure A1.

Table 8: Difference-in-Differences Estimates on the Probability of Employment, Log Hourly Wages and Hours Worked

2015 Expansion (BPS:12/17)	**Employed (Y/N)**		**Log(Hourly Wages)**		**Log(Hours Worked)**	
	Unweighted	IPW	Unweighted	IPW	Unweighted	IPW
Post-expansion x Treated	-0.014**	-0.002	-0.003	0.009	-0.004	0.006
	(0.007)	(0.009)	(0.007)	(0.010)	(0.007)	(0.011)
N individual-quarters	103490	103330	57390	57250	57480	57340
N Individuals	9330	9320	6000	5990	6010	5990
Event study p-value	0.42	0.63	0.65	0.77	0.62	0.52
Heterogeneous trend p-value	0.61	0.97	0.28	0.64	0.53	0.35

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The main estimates come from the DiD specification in Equation 10 in the main text. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort.

6.2.2. Heterogeneity by Race

Table 9 shows a similar pattern for the 2015 expansion. Hourly wages for Black borrowers increased by 5.4 percent and average weekly hours rose by 9 percent, both significantly larger than estimates for non-Black borrowers. The consistency of the wage finding across two distinct expansions and economic climates is notable, though the data do not permit a definitive account of the mechanism. Again, not shown, is an imprecise difference in job switching between Black and non-Black borrowers.

Table 9: Heterogeneous effect estimates by race, 2015 expansion

	Employed (Y/N)		**Log(Hourly Wage)**		**Log(Hours Worked)**	
	Black	Non-Black	Black	Non-Black	Black	Non-Black
Post-expansion x Treated	0.013	-0.005	0.051*	0.002	0.086*	-0.007
	(0.021)	(0.009)	(0.029)	(0.010)	(0.052)	(0.009)
N individual-quarters	18770	84560	7930	49320	7940	49400
N Individuals	1710	7610	890	5090	900	5100
Event study p-value	0.15	0.96	0.69	0.75	0.72	0.64
Heterogeneous trend p-value	0.83	0.87	0.46	0.44	0.95	0.35

Notes: *** $p< 0.01$, ** $p< 0.05$, * $p< 0.1$. Standard errors clustered at the individual level in parentheses. The estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort.

Similar explorations into heterogeneity of the main estimates by achievement and demographic characteristics did not reveal any substantive differences. The results by test scores and selectivity can be found in Appendix Table A3 and the differences by sex and marital status in Table A4. Overall effects on the monthly income distribution for this sample were also null (see Figure A2).

In this later 2015 expansion, further removed from recession and within a sample characterized mainly by more vulnerable students without degrees and with lower starting salaries, IDR expansion does not appear to have altered job search or earnings in meaningful ways. This could be related to the prominent IDR options already available at this time making REPAYE a weak treatment, but it could also be related to low awareness of IDR (Anderson et al., 2018) within a lower-achieving sample, and if so, represents a missed opportunity for a population that could have benefited from these policies.

7. Robustness

In this section I perform a series of robustness checks of alternative treatment definitions, PSLF introduction, and graduate school enrollment. The main employment results stand up to these checks.

7.1. Alternative Treatment Definition

The main estimates use non-borrowers as the comparison group. As a robustness check, I re-estimate the primary specifications using a borrower-only sample, defining treated borrowers as those whose predicted monthly payment would decrease under the new IDR plan and comparison borrowers as those whose payments would not.[11] The employment effect in this sample is 3.2 percentage points for the 2009 expansion, similar in magnitude to the preferred estimate of 2.1 percentage points. Results for all three outcomes and both cohorts are available in Appendix Table A5. Notably, the pre-trend tests reject parallel trends in the borrower-only employment specification (event study $p = 0.01$), which is evidence that heterogeneity across borrowers makes it difficult to identify a clean comparison group within the borrower population alone. This failure reinforces the case for the main strategy: non-borrowers have no theoretical reason to adjust labor supply in response to IDR expansion and exhibit strong pre-trend balance after re-weighting.

7.2. IDR enrollment

The main estimates capture the intent-to-treat effect of IDR expansion, regardless of whether borrowers actually enrolled. Take-up rates at this time were just 2 percent and 9 percent in 2009 and 2015, respectively, consistent with documented information barriers and program complexity (Cox et al., 2020; Anderson et al., 2018). Table 10 shows that borrowers who enrolled in IDR post-expansion saw a 3.3 percentage point employment increase in 2009, statistically indistinguishable from the 2.1 percentage point effect among non-enrollees ($p = 0.62$). Given that enrollees represent only 2 percent of borrowers, the aggregate employment effect cannot be driven by direct enrollment.[12] Together, these results support the interpretation that employment gains reflect the insurance value of IDR access rather than liquidity effects from actual enrollment.

[11] This test was a requirement to enroll in IBR in 2009, a designation officially referred to as showing "partial financial hardship."

[12] One logical alternative would be to implement a two-stage least squares (2SLS) design instrumental variables approach instrumenting enrollment with the expansion would require the exclusion restriction that expansion affected outcomes only through take-up. This conflicts directly with the insurance mechanism, which operates through option value regardless of enrollment, so I do not pursue this design.

Table 10: Heterogeneous effect estimates by post-expansion IDR enrollment and by expansion cohort

	Employed (Y/N)		Log(Hourly Wages)		Log(Hours Worked)	
	2009	**2015**	**2009**	**2015**	**2009**	**2015**
Enrolled in IDR post-expansion (Y/N)						
Post x Treated x No	0.021***	-0.006	-0.009	0.011	0.009	0.006
	(0.007)	(0.009)	(0.008)	(0.010)	(0.007)	(0.010)
Post x Treated x Yes	0.033	0.020	-0.003	0.002	0.004	0.022
	(0.023)	(0.017)	(0.033)	(0.017)	(0.029)	(0.021)
Yes = No (F, p-value)	(0.25, 0.62)	(2.61, 0.11)	(0.03, 0.87)	(0.29, 0.59)	(0.04, 0.85)	(0.73, 0.39)
Baseline rate (Yes)	0.02	0.09	0.02	0.09	0.02	0.09
N Observations	98030	103330	78910	57250	79320	57340
N Individuals	9350	9320	8140	5990	8160	5990

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. These estimates were generated by splitting the treatment group into two groups based on post-expansion enrollment in an IDR plan. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, a 3-category household income group, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter, exit cohort, and the same 3-category household income group.

7.3. Public Service Loan Forgiveness

The 2009 IBR expansion coincided with the introduction of PSLF, which incentivizes public service employment by promising early loan forgiveness after 10 years of qualifying payments. To rule out PSLF as an alternative mechanism, I conduct two tests. First, I compare employment effects for Direct loan borrowers who were PSLF-eligible against FFEL borrowers who would have been ineligible without consolidation of their loans. The effects are nearly identical at 2.3 and 2.0 percentage points respectively, inconsistent with PSLF driving the result. Second, I examine whether borrowers were more likely than non-borrowers to work in public service occupations post-expansion defined as education, healthcare, military, and social services. The DiD estimate on the probability of public service employment is 1.4 percentage points, but insignificant ($p = 0.35$). Neither test supports PSLF as a mechanism behind the employment effects.

7.4. Graduate School Enrollment

IDR expansion may have also influenced graduate school enrollment, which could mediate the employment result. Borrowers were marginally more likely to enroll in graduate school post-expansion (1.2 pp, $p < 0.1$). However, controlling for quarterly enrollment status leaves the employment estimate essentially unchanged (2.3 vs. 2.1 pp), and separate effects for graduate en-

rollees and non-enrollees are statistically indistinguishable (2.3 vs. 2.5 pp, $p > 0.1$). Overall this exercise suggests graduate school enrollment did not explain or diminish the probability of employment result from IDR expansion.

8. Conclusion

IDR has become a central feature of the federal student loan repayment system over the past fifteen years, growing from 5 percent of borrowers in 2012 to over a third by 2020. The recent elimination of SAVE and the introduction of the Repayment Assistance Plan under the One Big Beautiful Bill Act have substantially restructured the landscape, raising new questions about whether the labor market benefits documented here will persist under the new design. In this paper I have examined the labor supply and earnings effects of two major IDR expansions. The results offer a guide for understanding what features of repayment design are most consequential for borrower outcomes.

In the trough of the Great Recession, expanding IDR options raised employment among student loan borrowers with bachelor's degrees by 2.1 percentage points. This result cannot be fully explained by actual enrollment in IDR or by the coincident introduction of PSLF. Rather, the insurance value of IDR access played a significant role, particularly among borrowers with lower academic achievement and those who had difficulty making payments prior to the expansion. The employment gains were accompanied by a rightward shift in the earnings distribution of newly employed borrowers, consistent with the match quality prediction of the framework: by removing the delinquency clock that creates urgency to accept early offers, IDR allowed borrowers to search longer and accept better realizations from the wage distribution. These findings are consistent with prior evidence linking the IBR expansion to decreased default rates (Mueller & Yannelis, 2019b) and lower sensitivity to home price fluctuations, but add labor force participation and match quality to the list of documented benefits. The aggregate null on the intensive margin masks heterogeneity for Black borrowers who experienced hourly wage gains of 5–6 percent following both the 2009 and 2015 expansions. This finding holds across distinct economic climates and but its precise mechanism remains an open question for future research.

Smaller changes to eligibility and forgiveness terms in the 2015 REPAYE expansion did not elicit an aggregate response in labor force participation or earnings. At that time a robust IDR infrastructure was already available to most borrowers, possibly muting its effects. Especially in the BPS sample, which includes borrowers much less likely to have completed a degree and more

likely to have attended for-profit institutions, simply adding REPAYE to the suite of available plans was not enough to shift labor market outcomes. Anderson et al. (2018) show that fewer than half of student loan borrowers in 2016 had even heard of IDR plans, and the low and slow take-up reflects this. The null findings for 2015 represent a missed opportunity. Neither the liquidity nor the insurance benefits of IDR can take hold if borrowers do not know the plans exist.

The results of this paper speak directly to the changed policy environment. The IDR expansions studied here raised the insurance value of federal student loans by eliminating required payments and the threat of delinquency during spells of low earnings or unemployment. The framework in Section 3 identifies two mechanisms through which this matters: it lowers the cost of labor force entry for borrowers most exposed to non-payment risk, and it removes the urgency to accept early job offers, improving match quality. The new RAP modifies both margins. By replacing the poverty-line income protection with payments based on total AGI, RAP shifts the design logic from one that shields low-income borrowers from required payments toward one that requires a contribution at nearly all income levels. Whether this meaningfully constrains labor force entry depends on the magnitude of expected payments relative to the delinquency costs of the fixed plan. More consequential for the search channel may be RAP's progressive bracket structure, which introduces more salient income thresholds than the smooth taper of prior plans. If those brackets are salient enough to induce bunching just below rate-change points—the mechanism de Silva (2025) documents in Australia—the US system may begin to exhibit the intensive-margin distortions that have been absent to date. Whether RAP's design generates moral hazard, reduces the insurance channel, or both, is an open and important empirical question.

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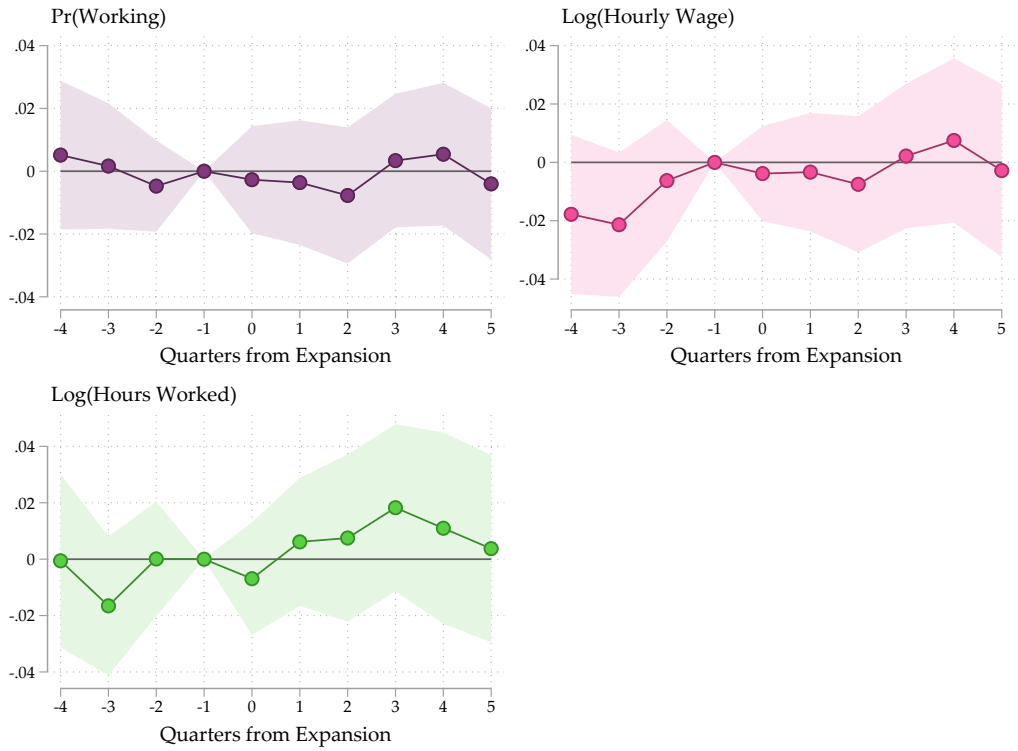
A. Additional Tables and Figures

Figure A1: 2015 Expansion Event Study Plots of Employment, Log Hourly Wage, and Log Hours Worked



Notes: The estimates are generated from linear regressions of the general form described in Equation 12 and are weighted by the IPW described in the main text. All figures include shaded regions for the 95 percent confidence interval of each estimate. Underlying sample comes from BPS.

Table A1: Intensive margin heterogeneous effect estimates by test score tercile and institution selectivity, 2009 expansion

	Test Scores				BA Institution Selectivity		
	Lowest Tercile	Middle Tercile	Top Tercile	No Scores	Low	Medium	High
Log(Hourly Wages)							
Post-expansion x Treated	0.015	-0.022	-0.008	-0.000	0.005	-0.008	-0.030
	(0.020)	(0.015)	(0.013)	(0.016)	(0.015)	(0.010)	(0.022)
N individual-quarters	8,760	24,540	31,880	13,720	15,610	52,370	10,930
N Individuals	910	2,520	3,340	1,370	1,580	5,390	1,180
Event study p-value	0.58	0.36	0.32	0.13	0.73	0.99	0.31
Heterogeneous trend p-value	0.22	0.08	0.49	0.15	0.73	0.82	0.08
Log(Hours Worked)							
Post-expansion x Treated	0.022	0.004	0.006	0.020	0.014	0.006	0.019
	(0.023)	(0.012)	(0.013)	(0.016)	(0.017)	(0.009)	(0.023)
N individual-quarters	8,800	24,670	32,030	13,820	15,680	52,640	11,000
N Individuals	910	2,520	3,340	1,380	1,580	5,400	1,180
Event study p-value	0.61	0.43	0.34	0.71	0.44	0.55	0.48
Heterogeneous trend p-value	0.65	0.14	0.69	0.94	0.23	0.22	0.37

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The main estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Test score terciles draw from the ACT national distribution from 2007: Low < 19, Middle $19 \leq TS \leq 23$, High > 23. Selectivity draws from the Barrons Competitiveness Index, where High selectivity includes the top 2 categories (most and highly competitive), Medium includes categories 3 and 4 (very and competitive), and Low includes all others including unclassified. All models include individual fixed effects and the interaction between quarter and exit cohort. Estimates were generated using the B&B cohort.

Table A2: Heterogeneous effect estimates by sex and marital status, 2009 expansion

	Employed (Y/N)		**Log(Hourly Wage)**		**Log(Hours Worked)**	
	Yes	No	Yes	No	Yes	No
Female (Yes/No)						
Post-expansion x Treated	0.021**	0.024**	-0.009	-0.007	0.008	0.011
	(0.009)	(0.010)	(0.011)	(0.012)	(0.010)	(0.010)
N individual-quarters	57510	40520	46150	32750	46390	32930
N Individuals	5480	3870	4770	3370	4780	3380
Event study p-value	0.55	0.73	0.77	0.34	0.63	0.43
Heterogeneous trend p-value	0.31	0.89	0.54	0.97	0.57	0.19
Married (Yes/No)						
Post-expansion x Treated	0.031***	0.017**	-0.008	-0.008	0.031**	0.001
	(0.012)	(0.008)	(0.012)	(0.010)	(0.014)	(0.009)
N individual-quarters	27900	70130	22580	56320	22680	56630
N Individuals	2620	6740	2260	5890	2260	5900
Event study p-value	0.40	0.99	0.18	0.92	0.23	0.49
Heterogeneous trend p-value	0.15	0.98	0.41	0.89	0.19	0.54

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors clustered at the individual level in parentheses. The estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort. Estimates are generated using the B&B sample.

Table A3: Heterogeneous effect estimates by test score tercile and institution selectivity, 2015 expansion

	Test Scores				Last Institution Selectivity		
	Lowest Tercile	Middle Tercile	Top Tercile	No Scores	Open Adm	Medium	High
Employed (Y/N)							
Post-expansion x Treated	-0.008	-0.012	-0.003	0.005	-0.001	0.026	-0.059**
	(0.017)	(0.017)	(0.024)	(0.014)	(0.010)	(0.026)	(0.028)
N individual-quarters	26610	21580	12400	42740	84800	11110	7430
N Individuals	2410	1950	1120	3840	7630	1000	680
Event study p-value	0.77	0.21	0.51	0.09	0.54	0.11	0.16
Heterogeneous trend p-value	0.21	0.26	0.43	0.08	0.19	0.06	0.03
Log(Hourly Wages)							
Post-expansion x Treated	0.029	0.009	-0.033	0.009	0.005	0.072**	-0.026
	(0.018)	(0.021)	(0.025)	(0.015)	(0.010)	(0.031)	(0.031)
N individual-quarters	13780	13120	7060	23290	46780	6100	4360
N Individuals	1490	1360	750	2390	4880	630	480
Event study p-value	0.39	0.99	0.95	0.62	0.89	0.11	0.76
Heterogeneous trend p-value	0.22	0.92	0.49	0.56	0.73	0.62	0.88
Log(Hours Worked)							
Post-expansion x Treated	0.003	0.011	-0.039	0.014	-0.004	0.086*	0.009
	(0.019)	(0.020)	(0.027)	(0.018)	(0.010)	(0.050)	(0.036)
N individual-quarters	13800	13140	7070	23330	46850	6110	4370
N Individuals	1490	1360	750	2400	4890	630	480
Event study p-value	0.57	0.28	0.98	0.57	0.49	0.28	0.63
Heterogeneous trend p-value	0.77	0.24	0.45	0.88	0.22	0.72	0.26

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The main estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Test score terciles draw from the ACT national distribution from 2011: Low < 19, Middle $19 \leq TS \leq 23$, High > 23. Selectivity draws from the variable, SELECT6Y, where High selectivity includes the top 2 categories, Medium includes all other schools with some selective admission process, and the lowest category is open admission institutions). All models include individual fixed effects and the interaction between quarter and exit cohort. Estimates were generated using the BPS sample.

Table A4: Heterogeneous effect estimates by sex and marital status, 2015 expansion

	Employed (Y/N)		Log(Hourly Wage)		Log(Hours Worked)	
	Yes	No	Yes	No	Yes	No
Female (Yes/No)						
Post-expansion x Treated	-0.006	0.001	-0.006	0.026*	-0.011	0.024
	(0.012)	(0.013)	(0.012)	(0.015)	(0.013)	(0.016)
N individual-quarters	57730	45600	30420	26830	30460	26870
N Individuals	5240	4080	3220	2770	3220	2770
Event study p-value	0.76	0.19	0.74	0.04	0.86	0.14
Heterogeneous trend p-value	0.53	0.40	0.69	0.80	0.54	0.48
Married (Yes/No)						
Post-expansion x Treated	0.029*	-0.014	0.019	0.004	0.023	-0.002
	(0.015)	(0.011)	(0.016)	(0.012)	(0.024)	(0.011)
N individual-quarters	27310	76020	15720	41530	15750	41590
N Individuals	2440	6880	1590	4390	1600	4400
Event study p-value	0.01	0.28	0.33	0.63	0.33	0.19
Heterogeneous trend p-value	0.76	0.83	0.49	0.96	0.42	0.15

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. The estimates come from the DiD specification in Equation 10 in the main text estimated separately for each category. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort. Estimates are generated using the BPS sample.

Table A5: Difference-in-differences estimates with borrower-only sample, by expansion cohort

	Employed (Y/N)		Log(Hourly Wage)		Log(Hours Worked)	
	2009	2015	2009	2015	2009	2015
Post-expansion x Treated	0.032***	-0.0184	0.006	0.003	0.013	0.005
	(0.010)	(0.0136)	(0.012)	(0.013)	(0.011)	(0.016)
N individual-quarters	39790	49230	29730	22360	29910	22400
N Individuals	3800	4500	3160	2480	3170	2480
Event study p-value	0.01	0.49	0.02	0.12	0.70	0.15
Heterogeneous trend p-value	0.05	0.71	0.12	0.56	0.71	0.41

Notes: *** p< 0.01, ** p< 0.05, * p< 0.1. Standard errors clustered at the individual level in parentheses. These estimates were generated using borrower-only sample, where the treated group are those borrowers whose monthly loan payment would have decreased under the new IDR plan and comparison borrowers are those who did not. Excluded from both groups are borrowers with positive household incomes above the full sample median. The event study parallel trend tests p-values come from Equation 11 and the heterogeneous trend test is described in subsequent text in the main paper applied to this smaller sample. Employed probability outcome model includes individual fixed effects and the interaction between quarter, exit cohort, an indicator for household income above 150% poverty line pre-expansion, and a pre-expansion period labor force participation indicator. The intensive margin columns include individual fixed effects and the interaction between quarter and exit cohort and an indicator for household income above 150% poverty line pre-expansion.

Figure A2: 2015 Expansion Effects on the Monthly Wage Distribution

Notes: Each bar and represents a DiD estimate along with a 90% confidence interval where the outcome is the probability of earning monthly wages in a given bin (e.g., $1,000) and the coefficient represents the difference pre- and post-IDR expansion between the treatment and comparison groups. Underlying sample comes from BPS.